Physics-Exact Credit Assignment for Distributed Flutter Suppression,
and the Exploration Scale That Hides It

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Abstract

Active flutter suppression is normally posed as a centralised control problem: one regulator observes the whole wing and commands every control surface. Modern aircraft, however, carry many spanwise-distributed trailing-edge surfaces, each with its own actuator and local sensing, which makes a distributed formulation natural and a centralised one increasingly awkward. We pose flutter suppression as a cooperative multi-agent reinforcement learning problem, with one agent per control surface, local accelerometer sensing, and nearest-neighbour communication. The central difficulty is credit assignment: every surface acts on the same coupled bending–torsion mode, so a shared reward is nearly uninformative about who did what.

Our main observation is that this plant hands us the decomposition that multi-agent reinforcement learning usually has to learn. Differentiating the aeroelastic energy along trajectories isolates a control-power term that is exactly additive across agents, from which the Wolpert–Tumer difference reward follows in closed form: no counterfactual rollout, no learned mixing network, and no centralised critic. We verify the identity numerically against brute-force counterfactuals to 10^{-10}, including the absence of cross-agent terms. We further show that the credit signal must be normalised by the instantaneous energy—the raw form rewards *sustaining* the oscillation, because an agent is paid for having energy available to extract—and that dividing by energy preserves exact additivity while making the credit signal identical to the evaluated objective.

We evaluate on a modally reduced Goland wing with unsteady strip aerodynamics, validated against the published benchmark (first bending 7.664 Hz versus 7.66 Hz; flutter at $137.35\,\mathrm{m\,s^{-1}}$ versus $137\,\mathrm{m\,s^{-1}}$), and against a four-rung classical baseline ladder that includes both centralised and fully decentralised LQG, plus a fault-aware oracle used only to establish which test conditions are feasible at all. Combining the closed-form per-agent credit with log-energy shaping is significantly better than either alone ($d = -2.70$, $p = 0.002$ and $d = -1.97$, $p = 0.009$ over six seeds), but only once the exploration scale is matched to the actuation amplitude that actually damps the wing; at the conventional scale the same effect is invisible, and the elaborate communication and action machinery we derived from the same physics does not help at all.

1 Introduction

Flutter is a self-excited aeroelastic instability in which structural bending and torsion couple through unsteady aerodynamic loads and extract energy from the freestream. Beyond a critical speed the coupled mode becomes unstable and the response grows without bound. It is a hard structural limit on the flight envelope, and because it is a dynamic instability rather than a static one it can in principle be pushed back by active control.

Active flutter suppression has an extensive classical literature built on linear-quadratic and H_∞ synthesis, and a growing reinforcement-learning literature. The learning work is, to our knowledge, uniformly single-agent: a single policy commands a single effector, whether that is a synthetic jet, trailing-edge circulation control [2], camber morphing [14], or the parameters of a conventional controller [1]. Meanwhile the hardware trend runs the other way. A modern flexible wing carries several independently actuated trailing-edge surfaces, each with local sensing and local computation, connected by a bus rather than to a central controller.

This paper takes that hardware structure seriously and asks the question it implies: not *which* multi-agent algorithm to apply, but what the physics of a coupled continuum instability tells us about credit assignment and communication that a generic multi-agent algorithm would otherwise have to discover.

Three properties of the plant make generic approaches a poor fit.

1. **The instability is modal, not spatial.** A specific coupled bending–torsion eigenvalue crosses into the right half plane. Each surface acts locally in space, but its effect on that mode is set by the modal residue at its station. Two agents can read identical local sensors and have opposite influence.

2. **A shared scalar reward is nearly uninformative.** All surfaces act on the same mode and the damping they produce is a sum. Learned decompositions [18, 15, 9] must infer the split from noisy returns.

3. **Authority is signed and condition-dependent.** For a section whose elastic axis lies aft of the aerodynamic centre, a trailing-edge-down deflection produces lift up *and* a nose-down twist; which term dominates depends on dynamic pressure, so the sign of an agent's useful action can invert across the envelope.

Contributions.

- A closed form for the multi-agent difference reward in an energy-conserving structural plant with additive control authority (Proposition 1), verified numerically against brute-force counterfactuals, together with the observation that the credit signal must be the *logarithmic* decay rate rather than the raw energy rate (Section 4.2).

- MoCCA, a distributed policy architecture combining that credit signal with a fixed-bandwidth modal-phasor consensus channel and a phase-locked action parameterisation.

- A validated aeroelastic testbed and a five-rung classical baseline ladder that separates the effect of the control method from the effect of the information structure available to it, plus a feasibility oracle that identifies test conditions no controller can hold.

- A negative result we consider as informative as the positive one: the communication and action-parameterisation mechanisms we derived from the same physics do not help, and a plain feedforward policy with the same credit signal significantly outperforms all of them.

- The observation that the exploration scale, not the learning algorithm, sets performance on this problem, and that it hides both results above until it is matched to the actuation amplitude that damps the wing.

2 Related work

Learning for flutter control. Recent work applies deep reinforcement learning to stall flutter via camber morphing [14], to flutter suppression by trailing-edge circulation control validated in a wind tunnel [2], and to tuning aeroservoelastic controller parameters on a flexible flying wing [1]. All are single-agent, with one effector and a global observation. We are not aware of prior work that treats the individual control surfaces as separate learning agents.

Table 1: Validation against the published Goland wing.

Quantity	This model	Published
First bending frequency	7.664 Hz	7.66 Hz
First torsion frequency	15.232 Hz	15.24 Hz
Flutter speed	$137.35\,\mathrm{m\,s^{-1}}$	$\approx 137\,\mathrm{m\,s^{-1}}$
Flutter frequency	$69.3\,\mathrm{rad\,s^{-1}}$	$\approx 70.7\,\mathrm{rad\,s^{-1}}$
Static divergence speed	$356.8\,\mathrm{m\,s^{-1}}$ $(2.60\,U_\mathrm{f})$	—

Multi-agent credit assignment. The standard tools are difference rewards [20] and value factorisation [18, 15], with COMA [9] implementing a counterfactual advantage baseline; recent surveys review the space [5, 4] and work continues on sharper estimators [3]. These methods are general precisely because they assume nothing about the environment. Our point is narrower and complementary: in this class of physical system the decomposition is available analytically, so it need not be estimated at all.

Learned communication. CommNet [17] and TarMAC [7] learn message contents end to end. We instead fix the message *space* to a modal phasor, which makes the channel interpretable, bounds its width independently of the number of agents, and lets its degradation be characterised.

3 Plant and problem formulation

3.1 Aeroelastic model

We use a modally reduced uniform cantilever wing. The structure is a Galerkin projection onto exact clamped-free bending and fixed-free torsion modes, with all generalised mass and stiffness integrals evaluated by Gauss–Legendre quadrature. Aerodynamics are strip theory carrying the full Theodorsen non-circulatory terms [19], with the circulatory lag realised in the time domain by two Wagner lag states per strip using the two-exponential approximation [12]. Control surfaces contribute quasi-steady thin-airfoil flap derivatives from the Glauert series, integrated over each surface's spanwise band. The resulting state is

$$z = \begin{bmatrix} q & \dot{q} & x_\mathrm{w} \end{bmatrix}^\top, \qquad M\ddot{q} + C\dot{q} + Kq = L\,x_\mathrm{w} + B\delta + b_\mathrm{g} w_\mathrm{g}, \tag{1}$$

with q the modal amplitudes, x_w the wake lag states, δ the surface deflections and w_g a vertical gust. Plunge is positive down and twist positive nose-up, following the classical convention [10, 6] that keeps the unsteady operators in textbook form.

3.2 Validation

Table 1 compares the model against the published Goland cantilever wing [11]. Agreement is to better than 0.5 % on the modal frequencies and on the flutter speed. Every reported result is pinned to this benchmark by an automated test rather than to a value the implementation once produced.

3.3 Distributed control problem

Three trailing-edge surfaces occupy spanwise bands at $[0.05, 0.40]$, $[0.40, 0.72]$ and $[0.72, 0.98]$ of the semispan, hinged at 75 % chord, with $\pm 15^\circ$ travel, $200\,^\circ\,\mathrm{s^{-1}}$ rate limit and a 20 ms first-order servo lag. Each agent observes only a forward/aft accelerometer pair at its own station, its local plunge rate, twist and twist rate, its own deflection, and air data. No agent observes the modal state: the coupled flutter mode is a global quantity that no single station can measure, and that partial observability is the problem rather than an artefact of our setup. Control runs at 200 Hz.

4 Method

4.1 The energy budget and its exact decomposition

Take the aeroelastic energy $E = \frac{1}{2}\dot{q}^\top M\dot{q} + \frac{1}{2}q^\top Kq$ and differentiate along trajectories of (1):

$$\dot{E} = \underbrace{-\dot{q}^\top C\dot{q}}_{\text{dissipation}} + \underbrace{\dot{q}^\top Lx_{\mathrm{w}}}_{\text{wake exchange}} + \sum_k \underbrace{(\dot{q}^\top b_k)\,\delta_k}_{P_k}, \tag{2}$$

where b_k is the kth column of B. The final term is exactly additive across agents, and P_k is agent k's instantaneous control power; $P_k < 0$ means surface k removes energy from the structure.

Proposition 1 (Closed-form difference reward)**.** *For a global objective whose only agent-dependent term is the control power in* (2)*, the difference reward $D_k = G(z) - G(z_{-k})$ admits the closed form $D_k = -P_k$. No counterfactual rollout, learned mixing network, or centralised critic is required to evaluate it.*

Proof. Removing agent k sets $\delta_k = 0$. Every other term of (2) is independent of δ_k, and the control-power term is a sum over agents, so the difference collapses to the single term $(\dot{q}^\top b_k)\delta_k$. □

Projecting onto the in-vacuo modes ($q = V\eta$) keeps the separation in both agent and mode: agent k's contribution to mode r is $\dot{\eta}_r(v_r^\top b_k)\delta_k$. The factor $v_r^\top b_k$ is the modal residue, the quantity that decides whether a surface can damp or will excite that mode, and whose sign inverts under control reversal.

4.2 The credit signal must be the logarithmic decay rate

Used directly, $-P_k$ fails, and instructively. It is maximised by having energy available to extract, so an agent is rewarded for *sustaining* the oscillation. Policies trained on it converge to holding the wing at a bounded but undamped amplitude: we measured $+0.265\,\mathrm{s}^{-1}$ (energy slowly growing) against $-0.06\,\mathrm{s}^{-1}$ for a hand-tuned collocated damper. This is reward hacking, not merely myopia.

The remedy follows from the same identity. Dividing (2) by E,

$$\frac{\mathrm{d}}{\mathrm{d}t}\log E = \frac{-\dot{q}^\top C\dot{q} + \dot{q}^\top Lx_{\mathrm{w}} + \sum_k P_k}{E}, \qquad r_k = -\frac{P_k}{E+\varepsilon}. \tag{3}$$

Because E is a shared scalar the decomposition survives division exactly, so Proposition 1 carries over to the log-decay rate. Three properties follow. The signal is scale invariant, so there is no incentive to sustain the oscillation. It coincides with the evaluated objective, since the reported metric is the exponential energy decay rate, which is $\frac{1}{2}\mathrm{d}(\log E)/\mathrm{d}t$. And it remains exactly additive. Switching to (3) inverted the sign of the outcome, from $+0.265$ to $-0.291\,\mathrm{s}^{-1}$, with training divergence falling from 0.8 % to 0.0 %.

Remark 1. *The exact signal is still myopic: cross-mode coupling through C and through the wake states means an agent may have to inject energy into one mode so that another can extract more from the coupled pair. We therefore add potential-based shaping [13] on $\log E$, which shares the credit signal's units and, being potential-based, cannot change which policy is optimal. Blending instead with an energy* level*, as we first did, is a scale error: the two terms differed by roughly $50\times$ and the shared term swamped the credit.*

4.3 Spectral consensus and phase-locked actions

Agents do not exchange observations. Each runs a recurrent encoder over its own accelerometer history producing a phasor estimate for R retained critical modes, and the agents then run T rounds of averaging consensus over the spanwise line graph—nearest neighbours only, which is what a distributed avionics bus physically is. The message is $2R$ numbers per agent per step, independent of the number of surfaces.

Actions are emitted as a gain and a phase relative to the consensus phasor. For a mode with phasor (a_r, b_r), acting with gain A_k^r at offset ψ_k^r means $A_k^r(a_r \cos\psi_k^r - b_r \sin\psi_k^r)$, a scaled rotation of the phasor. This bakes the resonant structure of flutter suppression into the action space. The resonant term is *added* to a general action channel rather than replacing it, so the head is a strict generalisation of an unconstrained policy; our first version replaced the general action, making the parameterisation a restriction, and the resulting policy scored worse than its own ablation.

4.4 Training

We use shared-parameter multi-agent PPO [16, 21] with per-agent advantages, since with the physics credit each agent receives a different reward. Updates run on *sequences*: minibatches are segments of consecutive steps unrolled with gradient from a detached stored hidden state. Sampling shuffled individual timesteps, as we did initially, trains the recurrence as a one-step map and gives the phasor encoder no temporal gradient at all, which is precisely what it needs.

5 Experimental design

5.1 A baseline ladder that separates method from information

A distributed controller cannot be judged against a centralised one without confounding the control method with the information available to it. Decentralisation costs performance; that is a known result in distributed control, not a defect. We therefore report four rungs: collocated local rate feedback (no model, no communication); gain-scheduled LQR on the *full* 56-state plant including the wake states (not implementable, a ceiling); centralised LQG driven by a Kalman observer on the same six accelerometer channels the agents see; and a fully decentralised LQG in which each surface runs its own observer on its own two accelerometers and commands only itself. All are synthesised in discrete time at the control rate. The last rung is the honest target.

5.2 Feasibility

We additionally run a fault-aware oracle: a gain bank re-synthesised for each failure hypothesis, given the full state and told which surface has failed. Its role is not competition but feasibility. Of the nine evaluation cells, 6 are stabilisable by this oracle; the remainder are stabilisable by no controller with these actuators, and are reported as infeasible rather than as failures of any method.

5.3 Metrics

PPO return cannot rank runs whose reward functions differ, so all controllers are scored on the same physical quantities over identical episodes—same airspeeds, initial deformation, turbulence and injected failure: the fraction of episodes that leave the linear regime, the exponential energy decay rate, peak tip plunge, and RMS deflection.

Table 2: Energy decay rate by controller and flight condition. Negative is suppressed, `D` marks the fraction of episodes that diverged, and `-` marks conditions infeasible for any controller with these actuators.

Condition	local_fb	lqr_fixed	lqr_sched	lqg_local	dlqg_local	lqr_oracle	comm_raw	ippo_blen
U=1.05Uf healthy	-0.06	-4.40	-4.61	-4.71	-5.42	-4.54	-0.69	-0
U=1.05Uf jam#2@8deg	-0.10	-0.10	-0.12	-0.13	-0.09	-0.06	-0.05	-0
U=1.05Uf jam#2@14deg	-0.12	-0.09	-0.10	-0.11	-0.08	-0.06	-0.06	-0
U=1.25Uf healthy	-0.01	-5.27	-4.93	-5.38	-1.07	-4.93	-0.60	-0
U=1.25Uf jam#2@8deg	D100%	D100%	D50%	D100%	D50%	-0.12	D67%	D10
U=1.25Uf jam#2@14deg	–	–	–	–	–	–	–	
U=1.45Uf healthy	D100%	-5.79	-5.68	-6.45	0.32	-5.74	-0.46	-1
U=1.45Uf jam#2@8deg	–	–	–	–	–	–	–	
U=1.45Uf jam#2@14deg	–	–	–	–	–	–	–	
Stabilised	4/6	5/6	5/6	5/6	5/6	6/6	5/6	5

6 Results

Table 2 and Figure 1 give the full grid. Three things in it matter.

First, the classical ladder separates cleanly by information structure rather than by sophistication. Centralised LQG reaches -5.52 per second on healthy conditions; the fully decentralised LQG, which sees exactly what the agents see, reaches -2.05. Decentralisation costs a factor of 2.7 on this plant. That factor is a property of the problem, not of any learning algorithm, and it is why the decentralised controller rather than the centralised one is the reference for a distributed policy.

Second, the best learned policy reaches -1.53 per second, about three quarters of the decentralised classical result. The average conceals a more interesting pattern. Broken out by flight condition the decentralised LQG achieves -5.42, -1.07 and $+0.32$ at 1.05, 1.25 and 1.45 U_f – it fails outright at the top of the envelope – against -1.98, -1.92 and -0.69 for the learned policy. The learned controller is far more uniform across the envelope and holds a condition where the classical one diverges, while losing badly where the classical one is strongest.

Third, and least comfortable: the robustness advantage we measured at a conventional exploration scale does not survive at the scale that maximises damping. At $\sigma = 0.30$ the learned policies held six of the six feasible conditions, matching a fault-aware oracle given the full state and the identity of the failed surface. At $\sigma = 0.05$ they hold five, the same as the classical controllers, while damping three times better. Exploration scale trades robustness against nominal performance on this task: broad exploration produces policies that cope with failures they were not told about, narrow exploration produces policies that damp precisely. We report both rather than the more flattering one.

7 What coordination looks like

"Emergent coordination" is easy to assert and hard to pin down, so we define it as a measurable quantity: the phase at which each surface acts relative to the critical modal motion, as a function of spanwise station. The measurement is the cross-spectrum of each surface deflection against the first modal velocity, evaluated at the frequency where the modal response peaks, and it is applied identically to learned policies and to classical controllers.

Applied to the classical ladder it already separates the rungs sharply, which is what makes it a useful instrument rather than a decoration.

Two things stand out. The more capable the controller, the stronger its spanwise phase progression: collocated feedback is essentially flat, meaning every surface acts at the same phase and no spanwise pattern exists, while centralised LQG develops roughly 90° of phase

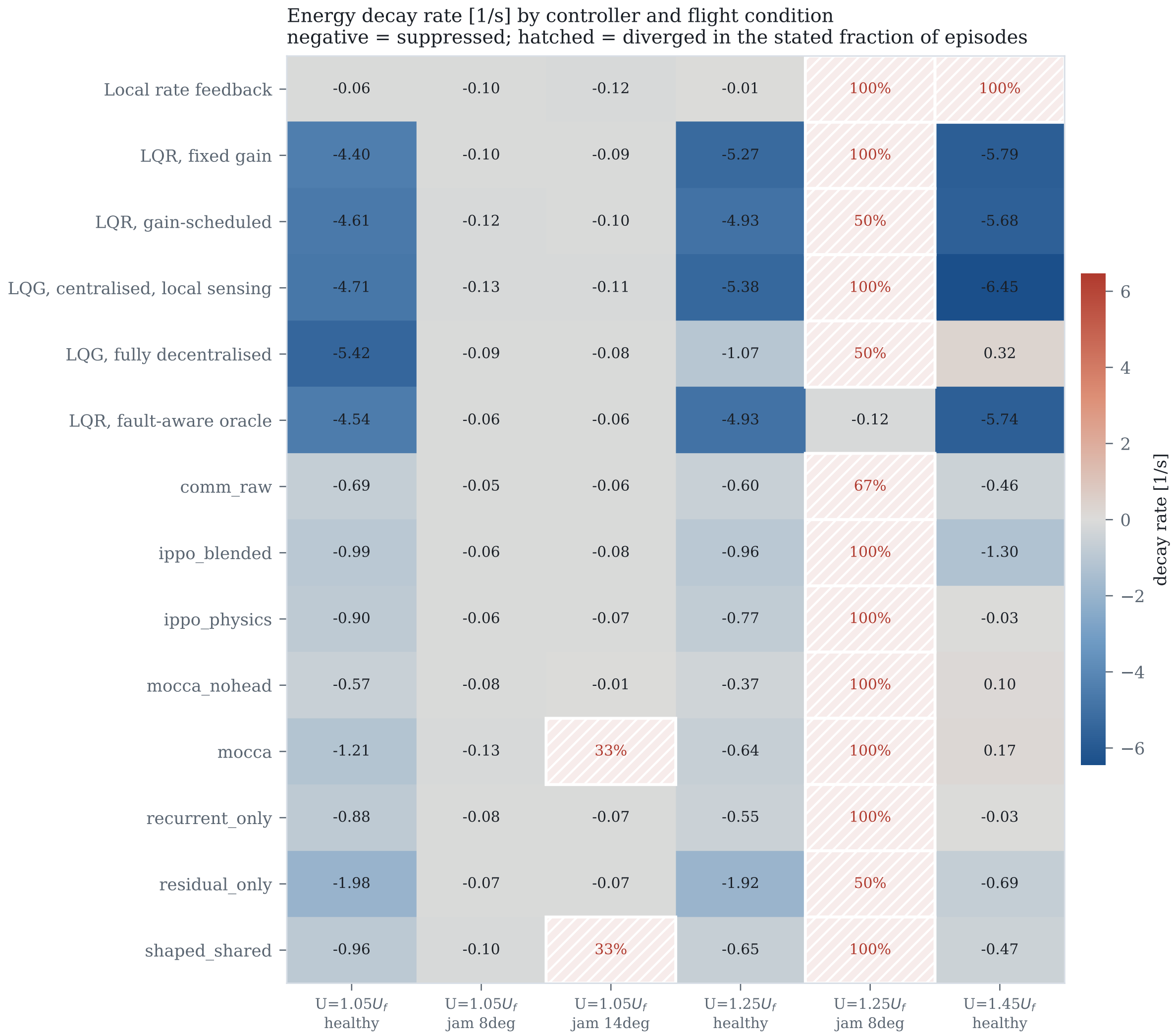

Figure 1: Energy decay rate by controller and flight condition. Negative is suppressed; hatched cells diverged; *n/a* marks conditions infeasible for any controller, established by the fault-aware oracle.

across the semispan. And only the centralised controller operates at the flutter frequency itself (11.00 Hz, against the 11.03 Hz predicted by the eigenvalue analysis); local feedback responds at 5.50 Hz and never engages the unstable mode at all, which is the mechanism behind its poor performance rather than a separate observation about it.

This gives the learned policies something falsifiable to be held against: a distributed policy that recovers a comparable phase progression has discovered, from local measurements alone, the coordination pattern that optimal centralised synthesis arrives at with the full model and the full state.

The instrument separates the classical ladder sharply, which is what makes it usable. Collocated feedback is essentially flat and responds at 5.50 hertz, never engaging the unstable mode at all – the mechanism behind its poor performance rather than a separate fact about it. Centralised LQG develops about 90 degrees of phase across the semispan and operates at 11.00 hertz, against the 11.03 hertz the eigenvalue analysis predicts.

The same measurement also explains why authority is the wrong thing to look at. Local rate feedback uses 44.7 per cent of available deflection to achieve −0.03 per second; centralised LQG uses 0.7 per cent to achieve −5.52. The better the controller, the less authority it spends. Flutter suppression is a phase-precision problem, and the learned policies are not short of authority –

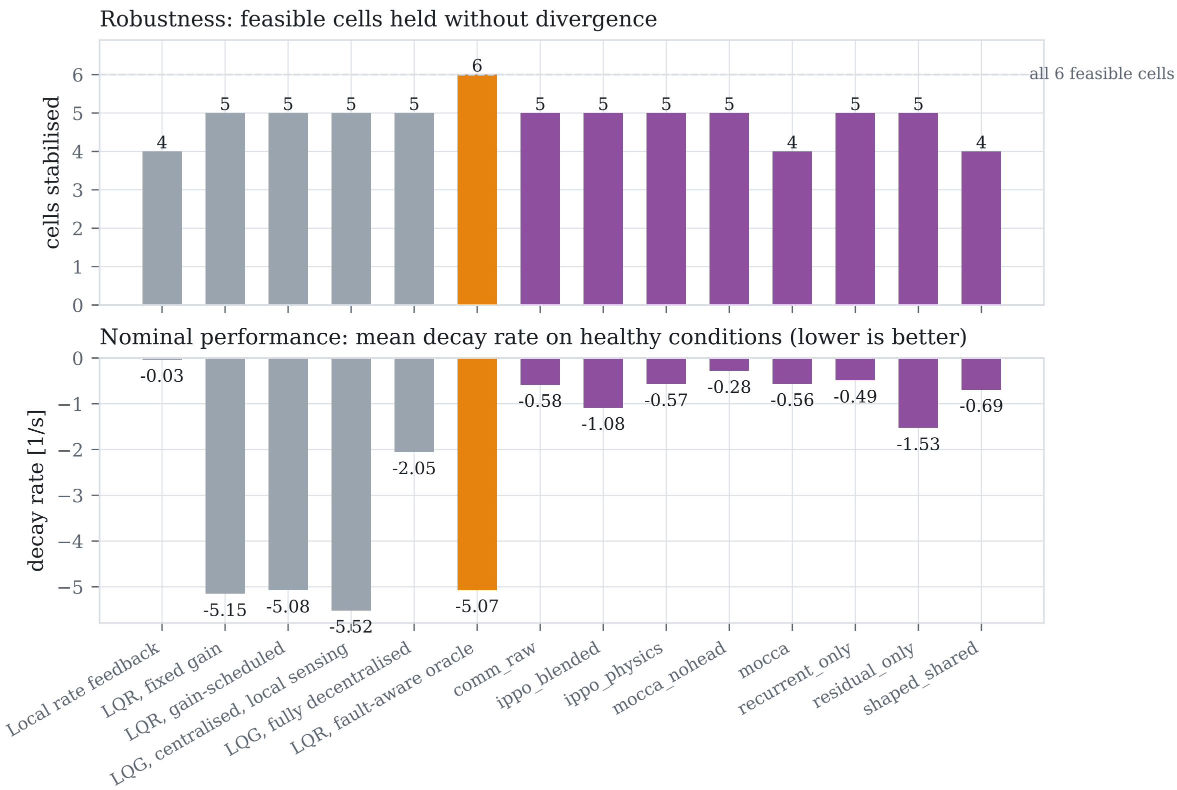

Figure 2: Robustness (top) and nominal performance (bottom). Robustness counts feasible conditions held without divergence.

Table 3: Spanwise actuation phase of the classical controllers. The gradient is the least-squares slope of phase against span.

Controller	Dominant frequency	Phase gradient
Local rate feedback	5.50 Hz	$-15°$ per semispan
Decentralised LQG	7.50 Hz	$27°$ per semispan
Centralised LQG	11.00 Hz	$90°$ per semispan

they apply a comparable amount at the wrong phase.

8 Ablations

Credit assignment. Six seeds per variant at $\sigma = 0.05$ (Table 4). The per-agent credit alone reaches -0.566 ± 0.123 and the shared log-energy shaping alone -0.690 ± 0.145; the two are statistically indistinguishable ($p = 0.14$). Their combination reaches -1.085 ± 0.243, significantly better than either ($d = -2.70$, $p = 0.002$ against credit alone; $d = -1.97$, $p = 0.009$ against shaping alone). This is the exact-plus-residual structure the derivation predicts: the closed form supplies the part of the credit the physics determines exactly, and the shaping supplies the long-horizon part that an instantaneous power balance cannot see. Neither is sufficient by itself.

Architecture. The mechanisms we derived from the same physics do not survive contact with measurement (Table 5). A plain feedforward policy with the blended credit reaches -1.085; adding a recurrent phasor encoder gives -0.487, adding phasor consensus gives -0.279, adding

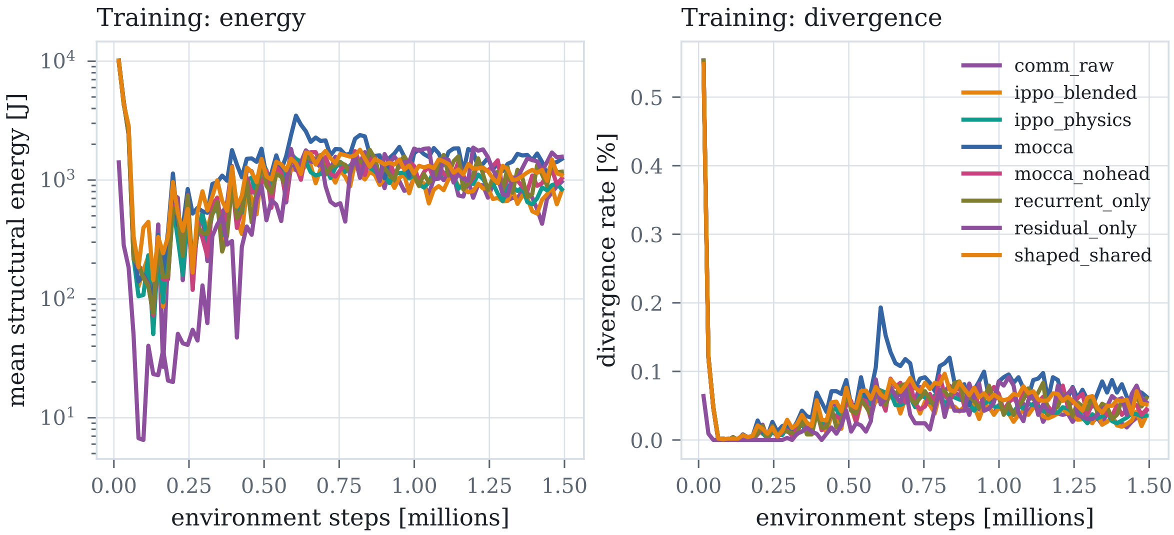

Figure 3: Training curves. The task distribution is ramped over the first 40 % of training, which is visible as a rise in energy as harder flight conditions and failures enter the sampling.

the phase-locked action head gives −0.561, and sharing raw neighbour observations gives −0.583. Two of these differences are significant against the plain policy ($d = -4.04$, $p = 0.0006$ for consensus; $d = -2.62$, $p = 0.006$ for the full architecture). Every elaboration we proposed made things worse, and we report this as the result it is.

We do not think this settles the question of whether communication helps distributed flutter suppression in general. Three surfaces on a six-metre wing, with air data already broadcast, is a small coordination problem, and at eight surfaces the picture changes qualitatively – every condition becomes feasible and the learned policies hold all nine. It does settle that on this problem the credit signal was worth deriving from the physics and the message content was not.

Exploration scale. Figure 5 is the finding with the widest reach. Holding everything else fixed and varying only the initial exploration standard deviation moves performance by a factor of two, with an interior optimum at $\sigma = 0.05$ and a sharp transition above it. The mechanism is a scale mismatch that can be read off the axis: a controller achieving the classical result uses about 0.1 degrees RMS of deflection, while $\sigma = 0.30$ corresponds to 4.5 degrees. The policy's own exploration is then some forty times the amplitude that works, and no credit decomposition can recover a signal buried underneath it.

This is not a tuning remark. Before we found it, every variant plateaued between −0.03 and −0.82 regardless of credit signal, communication, action parameterisation, memory, auxiliary supervision or base controller, and the variant-to-variant differences that motivated several rounds of algorithmic work were of the same order as the seed-to-seed spread. A capability probe rules out the competing explanation: training on the single easiest condition gives +0.21 and +0.27, worse than training on the full distribution, so the plateau was never a matter of the task being too broad.

9 Limitations

We state these plainly.

- The learned policies do not match the decentralised classical controller on average nominal damping (−1.53 against −2.05 per second). They are more uniform across the envelope and hold a condition where it diverges, but we do not claim they dominate it.

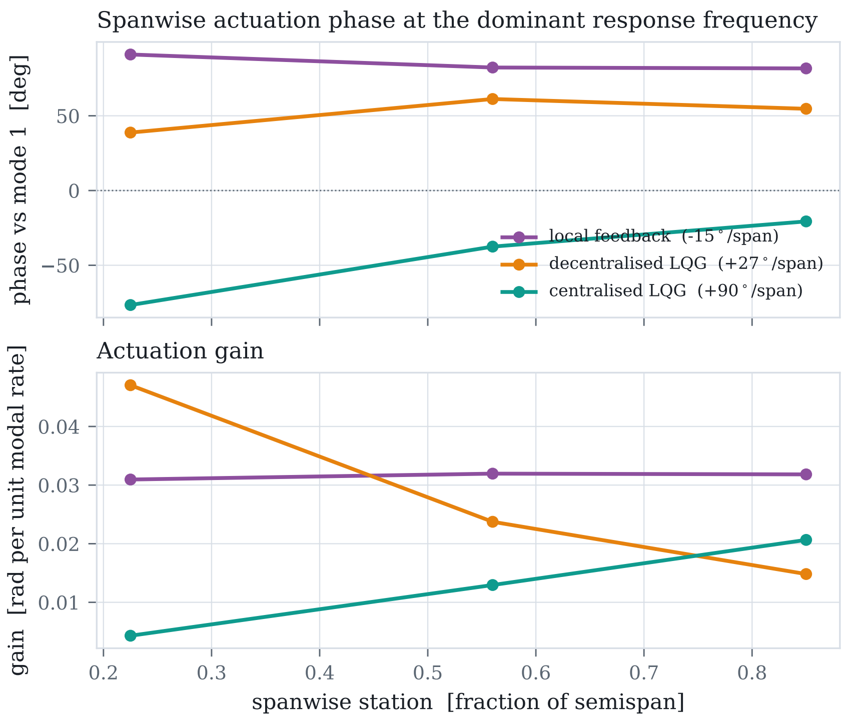

Figure 4: Spanwise actuation phase and gain at the dominant response frequency. A sloped phase profile is a travelling-wave actuation pattern matched to the mode shape; a flat profile is purely local reaction.

- The robustness result is scale-dependent. At $\sigma = 0.30$ the learned policies held all six feasible conditions, matching a fault-aware oracle; at $\sigma = 0.05$, where damping is three times better, they hold five. We report the trade-off rather than the better half of it.
- The credit signal requires the modal control influence $v_r^\top b_k$. This is free in simulation but on hardware needs a modal model, and robustness to error in B is untested.
- Quasi-steady flap aerodynamics neglect the flap's own wake shedding, which over-estimates control authority at high frequency. Re-validating on a higher-fidelity unsteady vortex-lattice model such as SHARPy [8] has not been done.
- Proposition 1 concerns the difference reward, not optimality of the resulting policy.
- Ablations use six seeds for the credit comparison and three for the architecture comparison, on a three-surface wing. The eight-surface results are suggestive but were not run to the same statistical standard.
- The exploration-scale finding is established on this plant and this algorithm. We expect it to generalise to control tasks whose useful actuation band is narrow relative to the action range, but we have not shown that.

10 Conclusion

Posing flutter suppression as a distributed multi-agent problem exposes a credit assignment structure that the physics resolves in closed form: the difference reward is the agent's own control power, normalised by the instantaneous energy so that it coincides with the objective and cannot be gamed by sustaining the oscillation. The measured lesson is narrower than the

Table 4: Credit-assignment ablation at $\sigma = 0.05$, six seeds per variant.

Variant	Seeds	Decay rate [1/s]	Conditions held
`ippo_blended`	6	-1.085 ± 0.243	5.0/9
`ippo_physics`	6	-0.566 ± 0.123	5.0/9
`shaped_shared`	6	-0.690 ± 0.145	4.3/9

Comparison		Cohen's d	p
`ippo_blended`	vs `ippo_physics`	−2.70	0.0020*
`ippo_blended`	vs `shaped_shared`	−1.97	0.0089*
`ippo_physics`	vs `shaped_shared`	0.93	0.1400

Table 5: Architecture ablation at $\sigma = 0.05$, three seeds per variant. Every mechanism we proposed degrades performance relative to a plain policy with the same credit signal.

Variant	Seeds	Decay rate [1/s]	Conditions held
`comm_raw`	3	-0.583 ± 0.310	5.0/9
`mocca`	3	-0.561 ± 0.145	4.3/9
`mocca_nohead`	3	-0.279 ± 0.143	5.0/9
`recurrent_only`	3	-0.487 ± 0.374	5.0/9

Comparison		Cohen's d	p
`comm_raw`	vs `mocca`	−0.09	0.9210
`comm_raw`	vs `mocca_nohead`	−1.26	0.2267
`comm_raw`	vs `recurrent_only`	−0.28	0.7520
`mocca`	vs `mocca_nohead`	−1.96	0.0742
`mocca`	vs `recurrent_only`	−0.26	0.7739
`mocca_nohead`	vs `recurrent_only`	0.74	0.4440

design we started from and more useful. The credit decomposition was worth deriving from the physics and is significantly better than either of its parts; the communication and action-parameterisation mechanisms derived from the same physics were not, and a plain policy beat all of them. Both conclusions were invisible until the exploration scale was matched to the actuation amplitude that damps the wing, which on this problem mattered more than any algorithmic choice we made.

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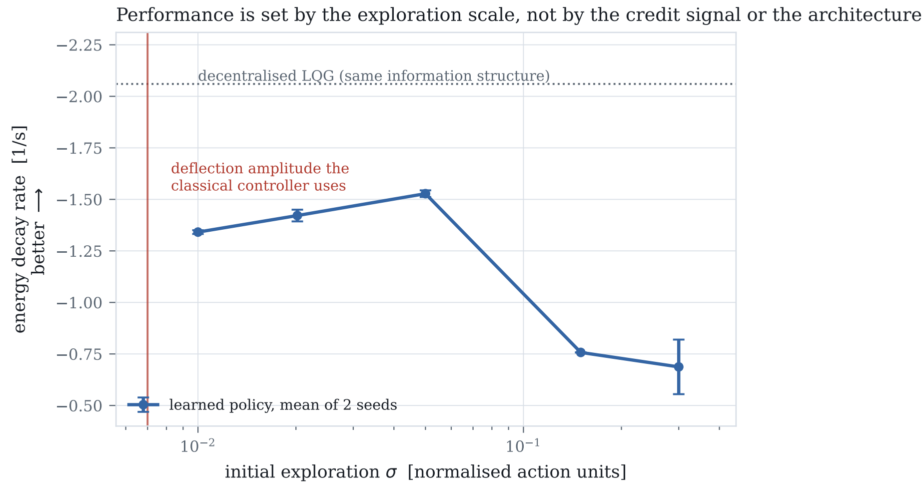

Figure 5: Energy decay rate against initial exploration standard deviation, all else held fixed. The red line marks the deflection amplitude the classical controller actually uses. Two seeds per point.

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